UNIT 2: SHAPES AND REGIONS

SESSION 9: REGION DESCRIPTORS

SLO 2: APPLY REGION DESCRIPTORS TO PERFORM SHAPE ANALYSIS AND OBJECT CLASSIFICATION IN IMAGES.

Feature Clue Game

A computer vision system extracted the following region features.

Identify the most likely shape.

Case 1

- Area is large
- Centroid is at the center
- Width ≈ Height

Predicted Shape: Circle

Case 2

- Area is moderate
- Height is much larger than width
- Centroid lies along the vertical center

Predicted Shape: Rectangle

Case 3

- Area is small
- Three major corners
- Centroid near geometric center

Predicted Shape: Triangle

Shape Investigator

A system detects three objects in an image.

Object	Area	Aspect Ratio	Eccentricity	Predicted Class
A	120	1 : 1	Low	Quadritarel
B	180	3 : 1	High	Polygon
C	90	1 : 1	Medium	Square

Question

Which region descriptor helped most in distinguishing Object B?

(3 : 1)